

Brazilian Olist E-commerce analysis

1. Business problem:

The company needs to understand the factors affecting sales decrease, customer retention in order to improve repeat purchases and long-term revenue :-

- Analyze the total sale for each month, with repeat rate for every months.
- Compare customer repeat rate between on time and delayed orders.
- Determine top products contributing to sales and gross profit.
- Investigate the profitability variance among all region.

2. Business context:

- Industry
 - E-commerce Retail
- Country
 - Brazil

3. Dataset Overview:

- Data source
 - Kaggle
- Time period
 - Sep 2016-sep 2018
- Records
 - 110197 orders
 - 99441 Customers
- Main Tables
 - Orders
 - Customers
 - Products
- Important columns
 - Order_id
 - Price
 - Order_date

- Delivery_date
- Product_id
- Product_category
- Customer_id
- Customer_state

4. KPI's :

- Total orders: 110197
- Revenue: 13221498.11
- AOV: 119.98
- Total customers: 99441

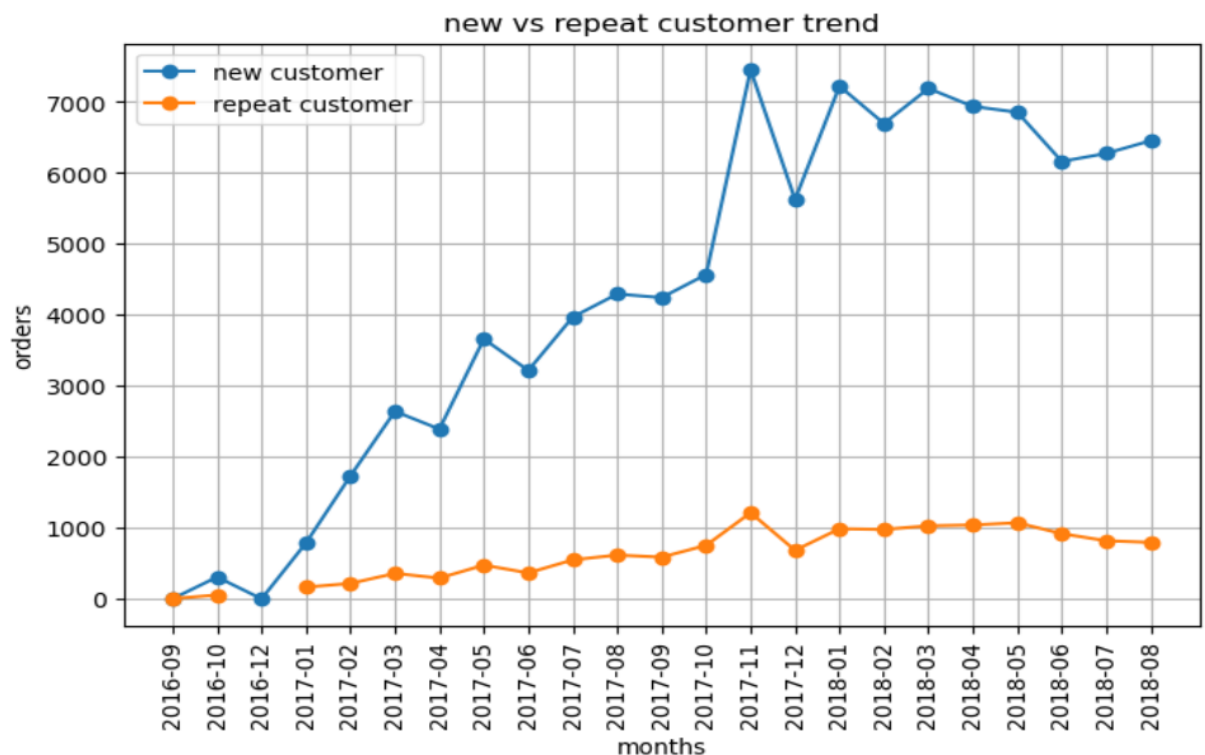
5. Feature engineering:

a. Create columns

- Delivery days – fetch days for each deliveries.
- Is delayed – flag delayed deliveries.
- State – region mapping.

6. Key Findings:

- New vs repeat customer trend



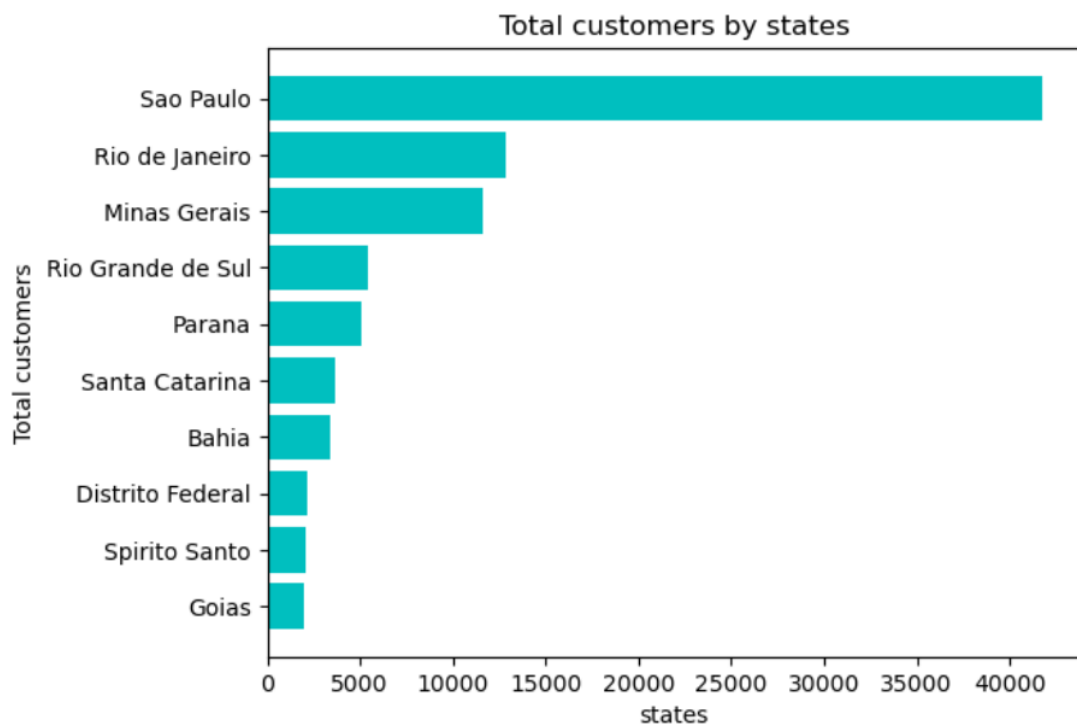
- The chart shows new customer has increases over the months and repeat customer line is inflated. This indicate that customers are not satisfied with the business.

- *Delivery delay impact*

delivery_type	total_customers	repeat_customers	repeat_rate
delayed	6535	510	7.804132
on time	92131	9293	10.086724

- Delayed deliveries reduced repeat customer rate, indicating the risk of increase on customer churn.

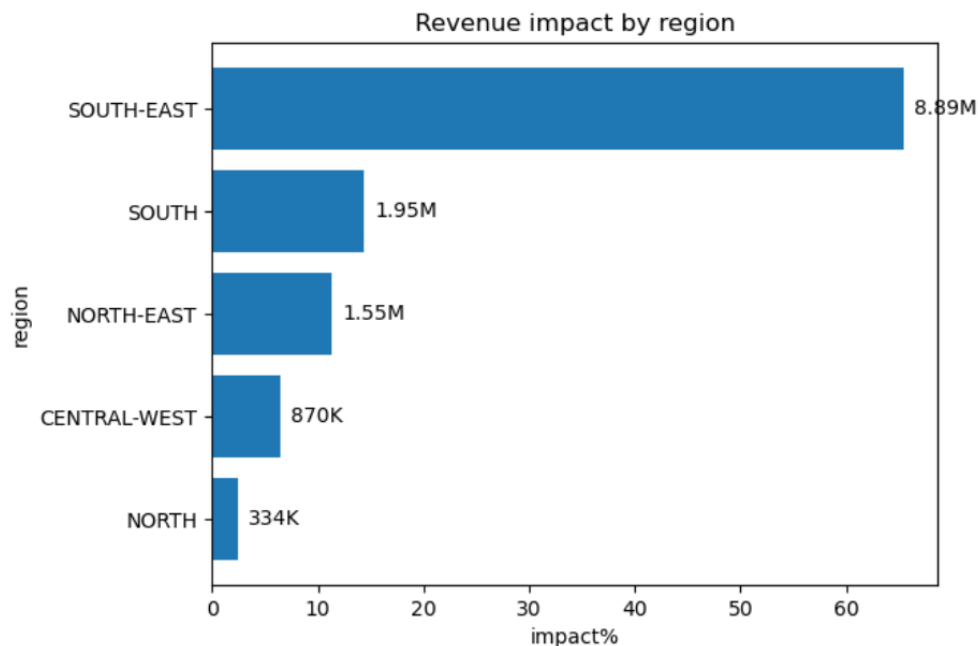
- *Top 10 states by customers*



- Sao Paulo has highest customer, increase marketing with customer loyalty campaigns to improve repeat purchase rate for top 3 states.

- Analyse delivery charges, pricing for remote and large population states.

- Regional sales analysis

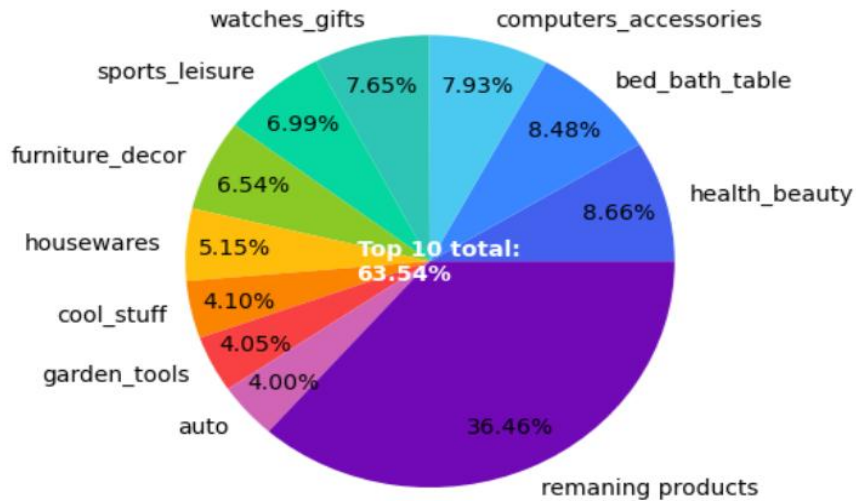


- Region South-East dominates with 65.38% impact of total revenue, increase marketing/inventory allocation to South-East to boost overall growth.
- North and Central-West combined has less than 10% impact on total revenue, investigate issue (logistics, competition) and create target campaigns to reduce the issue.

- Revenue contribution by product category

- Top 10 products contribute on revenue is 63.54 %
- The remaining products contribute only 36.46%, meaning they are not utilized effectively or may not be as competitive.

Revenue Contributions of top 10 products vs remaining products



7. Recommendations:

- Increase budget for top 10 products on marketing, inventory and lower budget for remaining product.
- Evaluate pricing and run offers for less purchased products.
- Increase marketing with customer loyalty campaigns to improve repeat purchase rate for top 3 states.
- North and Central-West combined has less than 10% impact on total revenue, investigate issue (logistics, competition) and create target campaigns to reduce the issue.

8. Business Impact:

By implementing these recommendations, the company can achieve customer retention, mitigate risks, and enhance overall revenue.